

Observations
sensor values,
positions + time

1
Fourier
embedding

CfC Memory
token attention,
causal scan dt

K,V

Branch Basis

2

**Cross-Attention
Readout**
isotropic distance bias

inner
product

Field
 u, v, p

Query
 (x, y, t_q, c)

Trunk Feature
Fourier + temporal
anchor, dt_to_query

Q

Trunk Basis

autograd

**SOAP +
Schedule-Free**
preconditioned
updates

update θ

optimize

3

Loss
AL continuity
constraint, sensor
MSE + NS
residual, GradNorm